

csvExportService.ts — Line-by-Line Reference

File Purpose

Browser-side CSV generation and download. Takes in-memory data rows and triggers a file download without any server round-trip. Exports two functions — one for survey data and one for route stats — that both follow the same download pattern.

Line-by-Line Breakdown

Lines 1–2 — Imports

```
import type { SurveyDataRow } from './surveyDatafetchService';
import type { RouteStatRow } from './routeDataFetchService';
```

Type-only imports so the module has no runtime dependency on either service. Only the shape of the row objects is needed here.

Lines 4–23 — exportToCsv

```
export function exportToCsv(filename: string, rows: SurveyDataRow[])
```

Builds and downloads a CSV of survey results (session ID, date, adults, children, total).

Lines	What happens
5	Defines the column headers: Session ID, Date, Adults, Children, Total
6–11	Maps each SurveyDataRow to a comma-joined string. numAdults/numChildren are coerced to numbers (defaulting to 0 if blank/null), and total is computed as their sum
13	Joins the header row and all data rows with newlines into a single CSV string
14	Wraps the string in a Blob with text/csv;charset=utf-8; MIME type
15	Creates a temporary object URL pointing to the blob
16–20	Creates a hidden <a> element, sets its href and download attribute, appends it to the DOM, clicks it to trigger the browser download, then removes it
22	Revokes the object URL to free memory

Lines 25–38 — exportRouteStatsToCsv

```
export function exportRouteStatsToCsv(filename: string, rows: RouteStatRow[])
```

Same download mechanism as exportToCsv , but for route stat rows (metric name + value).

Lines	What happens
26	Headers: Metric, Value
27	Each row is "metric label",value — the metric is wrapped in quotes to handle commas in label text
28–37	Identical blob → object URL → hidden link → click → cleanup pattern as above

Summary

Export	Input rows	CSV columns	Called from
exportToCsv	SurveyDataRow[]	Session ID, Date, Adults, Children, Total	survey/page.tsx Export button, export/page.tsx Survey Report
exportRouteStatsToCsv	RouteStatRow[]	Metric, Value	export/page.tsx Route Report

Both functions are synchronous and browser-only — they rely on `document` and `URL.createObjectURL`, so they cannot be called in a server context.